

Based on a thorough review of your manuscript and the scope of the **Journal of Air Transport Management (JATM)** – which emphasizes air transport economics, policy, operations, safety management, and sustainability – I recommend the following **additions and revisions** to strengthen your paper for submission and increase its chances of approval.

JATM values managerial relevance, empirical rigor, and clear links to industry decision-making. Your paper is strong methodologically, but it can be enhanced by adding explicit discussion on economic impact, broader industry applicability, and policy recommendations.

1. Add a Dedicated “Managerial Implications” Section (after Discussion)

JATM readers (airline/MRO managers, regulators, consultants) need actionable insights. Add a short section titled **“Managerial Implications for Air Transport Organizations”** with bullet points such as:

- **Cost-benefit justification for training transformation** – Present the calculated annual savings (\$1.2M from 28% error reduction) as an ROI metric (e.g., payback period, net present value).
- **Integration with existing SMS and ERP** – Provide a step-by-step checklist for MRO managers to embed Kirkpatrick levels into monthly quality reviews.
- **Audit-readiness strategy** – Show how the framework directly satisfies AS9100 §7.2 and CAA training documentation requirements, reducing audit non-conformances by 35%.
- **Sustainability reporting** – Link reduced rework (materials, energy, fuel) to ESG (Environmental, Social, Governance) metrics, relevant for airline sustainability reports.

2. Add a Short “Policy Recommendations” Subsection

JATM welcomes policy-oriented contributions. After the Conclusion, add **“Recommendations for Regulators and Industry Bodies”**:

- Propose that CAAs (e.g., FAA, EASA, ANAC) require Kirkpatrick-based training evaluation as part of SMS acceptance criteria.
- Suggest that IATA and SAE develop a standardized guidance document for MRO training evaluation based on this framework.

- Recommend that aviation safety regulators mandate annual reporting of training effectiveness metrics (Levels 3 & 4) as part of safety performance indicators.
- ### 3. **Strengthen the Literature Review with JATM-Published Papers**
- Add references to **recent JATM articles** to show alignment and build on the journal’s existing conversation. Suggested citations (examples – please verify):
- **“Airline maintenance costs and training efficiency”
 - **“Sustainability in air transport MRO”
 - **“Human factors and economic impacts in aircraft maintenance”

Action: Search JATM archives for keywords: “maintenance training,” “human error cost,” “MRO sustainability.” Add 3–5 such references.

4. **Add a Comparative Table of Key Results

JATM values quantitative summaries. In the Results section (4.4 or after), insert **Table 1: Summary of Operational and Sustainability Outcomes

Metric	Before Implementation	After Implementation	Improvement
Training failures (%)	Baseline (internal)	-28%	28% reduction
Regulatory non-conformances	Baseline	-35%	35% reduction
Quality costs (USD/year)	Estimated \$X M	\$Y M	\$Z M savings
Customer satisfaction (scale 0–5)	[Insert pre-data]	[Post-data Fig 10]	+Δ%
Material waste (kg or USD)	[Estimate from CoQ]	Reduced	Avoided waste

If pre-implementation data is not shown in current paper, estimate from historical averages mentioned (e.g., \$189,500 per warranty claim). This table will be powerful for JATM reviewers.

5. **Elaborate on the “Economic Analysis” Subsection

Currently Section 4.4 (“Impact of ineffective training on Costs of Quality”) is descriptive. Add a **quantitative economic model**:

- **Internal Failure Costs** – Rework labor hours × hourly rate, scrapped components × average cost.
- **External Failure Costs** – Warranty claims, concessions, shipping costs.
- **Prevention & Appraisal Costs** – Cost of implementing the Kirkpatrick framework (e.g., software, instructor training) – then show net savings.

Present a simple **ROI calculation**:

>
$$\text{ROI} = (\text{Total savings} - \text{Total implementation cost}) / \text{Total implementation cost}$$

This aligns with JATM’s emphasis on managerial decision support.

6. **Add a “Limitations and Future Research” Table**

JATM values transparency. Replace the prose limitations with a **clear table** (or keep prose but add a structured box):

Limitation	Mitigation in Study	Future Research Direction
Single case study (one MRO facility)	In-depth triangulation	Multi-site replication across regions
Inability to fully isolate training effects from other variables (mentioned in Lit Review)	Used SHAP attribution	Longitudinal RCT or quasi-experimental design
Self-reported behavior data	Dual questionnaires + objective quality metrics	IoT-based behavioral tracking
Generalizability to smaller MROs	Provides scalable version (e.g., simplified Level 3)	Adapt framework for low-resource settings

7. **Revise the Abstract for JATM Style**

JATM prefers abstracts that state **practical relevance** and **economic outcomes** upfront. Suggested revision (changes in *italics*):

> “Within high-risk industries like aviation maintenance, safety and sustainability are intrinsically linked ... *This study provides a replicable framework for MRO managers and*

regulators, demonstrating that systematic training evaluation generates annual quality cost savings of over \$1.2M while reducing regulatory non-conformances by 35%. The findings support policy recommendations for integrating training effectiveness metrics into Safety Management Systems.*”

8. **Add a Graphical Conceptual Model (Figure 0 or in Introduction)**

Create a **new figure** showing how Kirkpatrick’s four levels map directly to:

- AS9100 clauses (e.g., Level 4 → §9 Performance Evaluation)
- CAA SMS pillars (e.g., Level 3 → Safety Assurance)
- Sustainability outcomes (waste, energy, CO₂)

This visual will appeal to JATM readers and be suitable for the journal’s graphical abstract.

9. **Minor but Important – Clarify “Training Failures”**

In multiple places you mention “training failures reduced by 28%.” Define clearly: **“Training failure = an instance where a maintenance technician, after completing training, commits a documented procedural error directly attributable to a knowledge or skill gap identified in post-training evaluation.”** Add a short operational definition in Section 3.2.

10. **Check and Update References to JATM Format**

Ensure all references follow JATM’s style (likely Elsevier – author-date with journal name in italics). Add DOI links where available. Specifically, verify that the Chen et al. (2023), Sousa et al. (2025), and Yang & Park (2024) papers are indeed from JATM or similar high-impact journals – if not, replace with equivalent JATM papers from 2022–2025.

Summary Checklist for Submission to JATM

- [] Add **Managerial Implications** section (explicit ROI, ESG, audit readiness)
- [] Add **Policy Recommendations** subsection (targeting CAAs, IATA, SAE)
- [] Insert **Table of Key Operational & Sustainability Outcomes** (quantified)
- [] Expand **economic analysis** with ROI calculation and cost categories
- [] Add **Limitations & Future Research Table**

- [] Revise **abstract** to emphasize economic and policy impact
- [] Include **conceptual figure** linking Kirkpatrick → AS9100 → SMS → sustainability
- [] Define “training failure” operationally
- [] Add 3–5 recent **JATM references** to literature review
- [] Verify all citations follow JATM formatting (Elsevier)

These additions will align your strong case study with JATM’s focus on **air transport management, policy, and operational economics**, significantly improving your chances of acceptance. Good luck!